Jake Golden

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EDUCATION

Duke University – Pratt School of Engineering Expected May 2027
Bachelor of Engineering in Electrical and Computer Engineering, Biomedical Engineering *Durham, NC*
Cumulative GPA: 4.00/4.00; Dean's List with Distinction (2024, 2025, 2026).

EXPERIENCE

Analog Design Intern May 2026 – Aug 2026
National Instruments (Emerson) *Austin, TX*
- Designed schematic and board layout for precision LCR/SMU meter characterization PCB, implementing low-leakage switching architecture in guarded multiplexed signal chain; led schematic/layout reviews with cross-functional partners
- Authored 15+ V&V tests for LCR/SMU meter, automated instrumentation control and data acquisition/processing using LabVIEW, compiled results in formal report.

Product Development Intern May 2025 – Aug 2025
Stryker *Mahwah, NJ*
- Developed 9 prototypes for novel implant; created design inputs to address unmet user needs and models in PTC Creo.
- Verified GD&T equivalency to legacy devices, supporting design review and regulatory documentation.
- Automated generation of digital surgical template files using PTC Creo/MATLAB, saving 10+ hours of work.

Computational Neuroscience Researcher Jan 2024 – Present
Duke University Medical Center – Brain Stimulation Engineering Lab *Durham, NC*
- Programmed finite element analysis (FEA) ECT/TMS brain stimulation models in MATLAB.
- Developed automated pipelines (MATLAB/Bash) integrating MRI image processing, mesh generation (4M+ elements), and TMS coil optimization for high-throughput, patient-specific FEA simulations (~50/week).

Mechanical Engineering Intern Sep 2022 – Jun 2023
SIMS Pump Valve Company *Hoboken, NJ*
- Created ISO-standard CAD models/drawings using surface modeling techniques in SOLIDWORKS.
- Built standardized part library (300+ parts) for efficient modeling of pump systems in future projects.
- Performed tolerance verification on manufactured pump parts using calipers, micrometers, and 3D scanning.

PERSONAL PROJECTS | *Visit jake-golden.github.io for more*

MATLAB Pulse Oximeter
- Designed single-supply analog front-end conditioning circuit (TIA + active filtering + biased gain) converting photodiode current to PPG voltage waveform.
- Implemented custom bidirectional Arduino-MATLAB serial protocol with fixed-length packets, start-byte framing, and checksum validation.
- Developed MATLAB DSP (filtering, amplitude extraction, R-ratio-to-SpO_2 calibration) and GUI layer (MATLAB app) with live plotting, LED control, and diagnostics console.

PID-Controlled Self-Balancing Robot
- Built two-wheeled self-balancing robot with Arduino, IMU sensors, and DC motors.
- Implemented I^2C communication between sensors and motor drivers on custom-designed PCB.
- Iteratively tuned PID control parameters to achieve stable balancing.

SKILLS

Analog/Digital Circuit Design: LTSpice, KiCad, Siemens Xpedition, op-amp circuit design, filter design, basic CPU design, combinational/sequential Boolean logic, power electronics and power supply design
Programming: MATLAB & Simulink, Python, C, Java, LabVIEW
Mechanical Design: SOLIDWORKS, PTC Creo, Autodesk Fusion, 3D printing (FDM), laser cutting
Applied Mathematics: Signal processing/filtering algorithms, control theory, Fourier analysis, transmission line theory, electromagnetics, hypothesis testing, inferential statistics

ACTIVITIES & INTERESTS

Duke University football student manager, classical/jazz piano, photography, college basketball enthusiast, math tutor